

## Responses to Requests from the Technical Task Group

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*This document responds to requests from the Technical Task Group of the Small Pelagic Scientific Working Group, particularly relating to investigations of stock-recruitment relationships for the Warm Temperate Sardine.*

*Keywords: parameter estimation, stock-recruitment relationships, sardine, South Africa*

### Introduction

This document responds to requests from the Technical Task Group (TTG) of the Small Pelagic Scientific Working Group for specific tests and output, particularly relating to (i) parameter values and the AD Model Builder Hessian-based estimates of standard deviation and (ii) investigating alternative stock-recruitment relationships for the Warm Temperate Sardine.

### Estimated parameter values and standard errors at the joint posterior mode

Table A1 lists the parameter values and their Hessian-based standard errors at the joint posterior mode for the updated base case model ( $S_B$ ) as defined at the September 2025 TTG meeting (de Moor 2025b), with recruitment estimated independently each year and natural mortality assumed to be time-invariant.

### Fitting stock recruitment relationships after conditioning

Figure 1 shows the Hockey-Stick and Beverton Holt stock recruitment relationships estimated after conditioning for  $S_B$ . As previously discussed by de Moor (2025a) and the TTG, a number of near-zero CTS recruitments over a wide range of spawner biomass estimates results in stock recruitment relationships with very low asymptotes. These may be unrealistic.

### Comparing median recruitment and associated standard deviation between different spawner biomass ranges

As a method to compare if median recruitment, and in particular the variability in this recruitment, changes between 'middle' spawner biomass levels and 'high' spawner biomass levels for WTS, the following was done:

- i) The 3 peak recruitments from November 2000-2002 were excluded.
- ii) The 6 recruitments corresponding to the lowest 6 spawner biomass estimates were grouped and set aside; in future these may be assumed to correspond to the descending limb of a Hockey-Stick stock recruitment relationship.
- iii) The 4, 5, 6, 7 or 8 recruitments corresponding to the 4, 5, 6, 7 or 8 highest spawner biomass estimates, respectively, were separated from the remaining recruitments. Constant recruitment,  $a$ , was estimated for each of these sets by minimising  $-\ln L = \sum_i (\ln(a) - \ln(R_i))^2$  and calculating the associated standard deviation for each set.

The different ranges are shown in Figure 2, while Table 1 gives the results. There is little difference in the estimated median recruitment or standard deviation between the 'middle' and 'high' spawner biomass ranges if the 'high' range consists of the top 6, 7, or 8 values. If the 'high' range consists only of the top 4 or 5 values then the median recruitment is estimated

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to be higher and the standard deviation lower for the ‘high’ range. AIC model selection does not prefer using only the top 4 or 5 values.

#### Estimating a stock recruitment relationship during conditioning for CTS only

The TTG noted that results presented thus far indicate that a Beverton Holt stock recruitment relationship can be reliably estimated for CTS. Model  $S_{CTS}$  differs from  $S_B$  by estimating a Beverton Holt stock recruitment relationship during conditioning for CTS, assuming  $\sigma_{r,CTS}^2 = 5$ , while estimating WTS recruitment independently. The fit to the data is slightly worse than that for  $S_B$  (Table 2), but the recruitment deviates are likely more robustly estimated (Table A1, Figure 3).

#### Estimating a Hockey Stick stock recruitment relationship during conditioning for WTS

Model  $S_{CTS}$  was then extended to estimate a Hockey Stick stock recruitment relationship for WTS during conditioning ( $S_{HS}$ ). This test was run assuming  $\sigma_{r,WTS}^2 = 5$  for all years and with the peak recruitments estimated about a different median value ( $c$ ) to the Hockey Stick relationship. As for  $S_{CTS}$ , the fit to the data is slightly worse (Table 2), but this option has the advantage of stock recruitment relationships estimated for both stocks during conditioning.

#### References

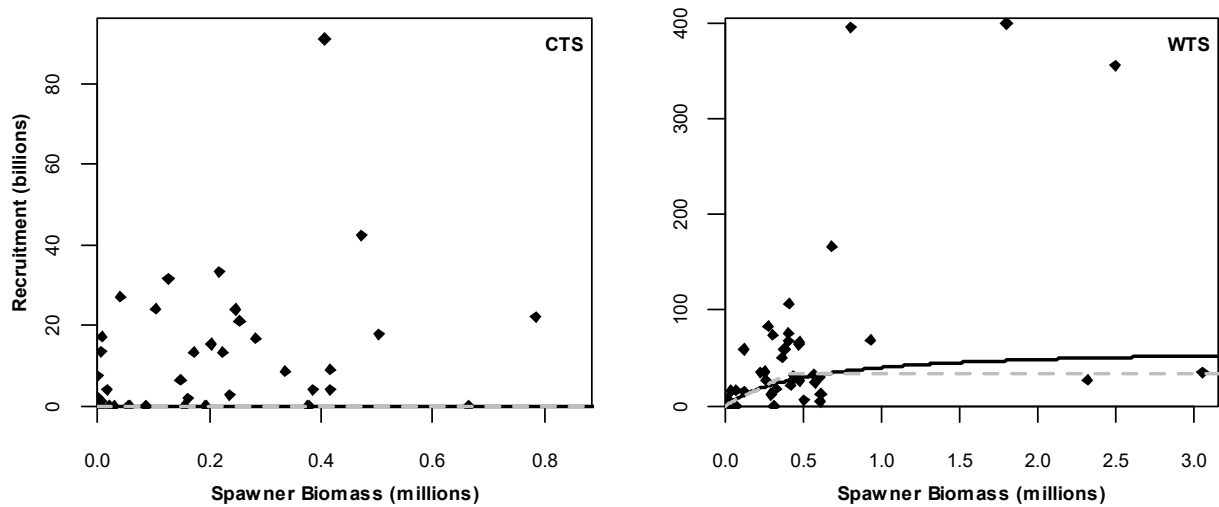
- de Moor CL. 2025a. Investigating the impact on the South African sardine assessment of estimating Beverton Holt stock recruitment relationships during conditioning. DFFE: Branch Fisheries Document FISHERIES/2025/MAY/SWG-PEL/28.
- de Moor CL. 2025b. Exploring further key uncertainties about the sardine assessment. DFFE: Branch Fisheries Document FISHERIES/2025/SEP/SWG-PEL/49.

**Table 1.** The estimated median recruitment corresponding to ‘middle’ and ‘high’ spawner biomass levels, where the ‘high’ range includes the highest 4, 5, 6, 7 or 8 estimates. The ‘middle’ points are denoted by red circles while the ‘high’ points are denoted by green circles in Figure 2. The blue circles in Figure 2 denote points which are either ‘high’ or ‘middle’ depending on how many are used in the ‘high’ range. The AIC is calculated as  $2N - 2\ln L_{mid} - 2\ln L_{high}$

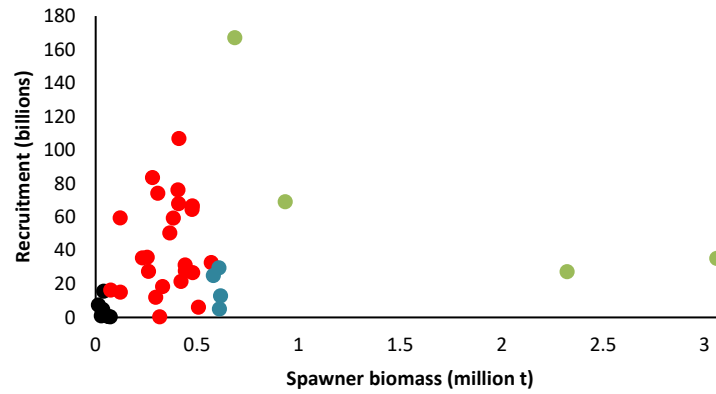
| Cut-off for<br>‘high’ range | $-lnL$    |      | $a$    |      | SD     |      | AIC  |
|-----------------------------|-----------|------|--------|------|--------|------|------|
|                             | SSB range |      |        |      |        |      |      |
|                             | Middle    | High | Middle | High | Middle | High |      |
| Highest 4                   | 42.09     | 1.97 | 28.9   | 57.6 | 1.23   | 0.70 | 90.1 |
| Highest 5                   | 38.46     | 3.77 | 28.9   | 42.7 | 1.19   | 0.87 | 86.5 |
| Highest 6                   | 33.84     | 7.62 | 28.9   | 29.8 | 1.14   | 1.13 | 84.9 |
| Highest 7                   | 33.84     | 7.62 | 28.9   | 29.8 | 1.16   | 1.04 | 84.9 |
| Highest 8                   | 33.81     | 7.65 | 29.1   | 29.8 | 1.19   | 0.98 | 84.9 |

**Table 2.** The contributions to the objective function from likelihood (equations A29 – A33 of de Moor 2025c) and prior components for  $S_B$  and  $S_{CTS}$  together with some key parameter estimates at the joint posterior mode.

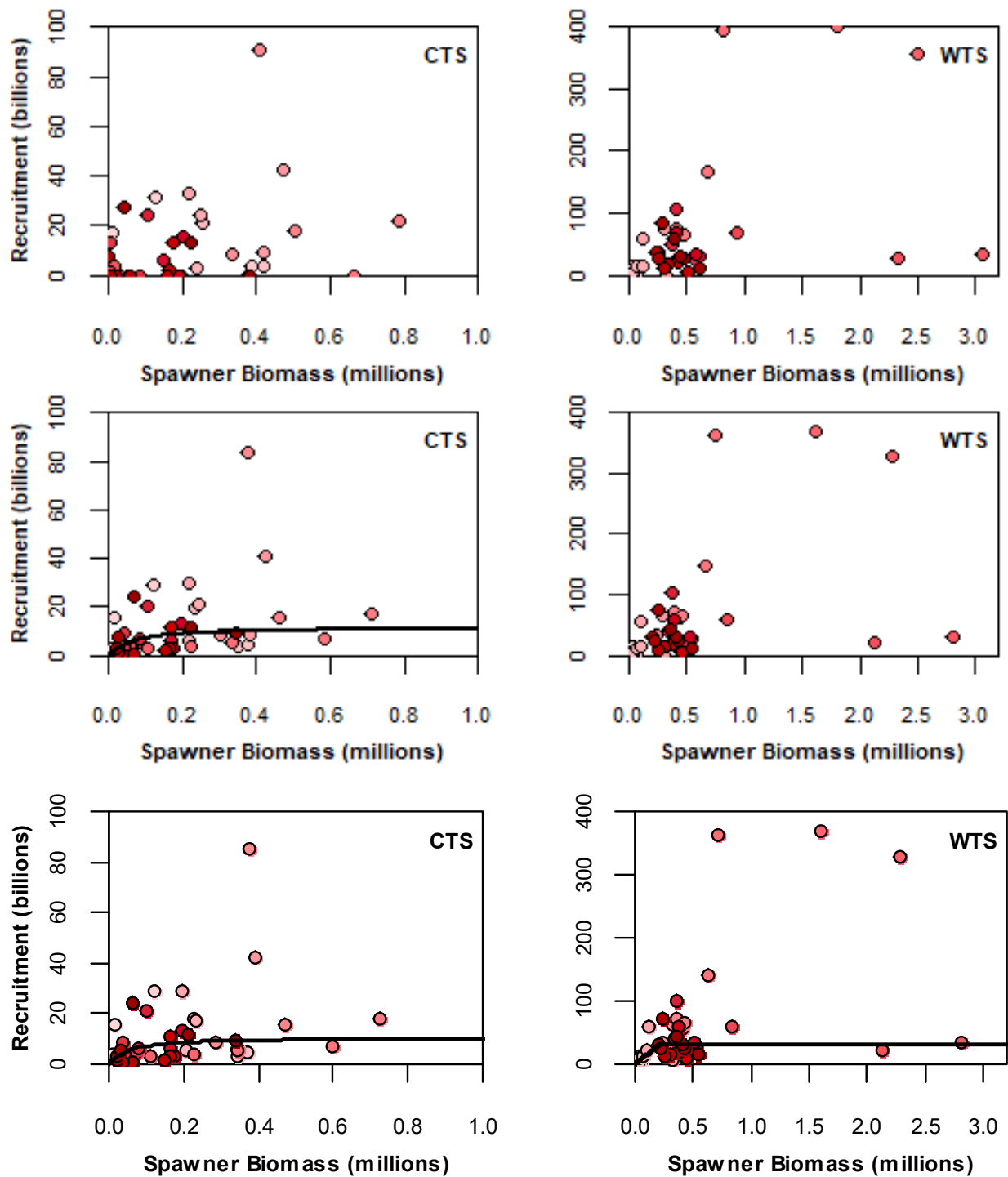
|                    |                      | $S_B$  | $S_{CTS}$ | $S_{HS}$ |
|--------------------|----------------------|--------|-----------|----------|
| Objective function |                      | -837.0 | -760.0    | -685.7   |
| log likelihood     | $-\ln L$             | -901.2 | -898.4    | -897.4   |
|                    | $-\ln L^{Nov}$       | 41.5   | 41.7      | 41.7     |
|                    | $-\ln L^{rec}$       | 30.9   | 31.2      | 32.1     |
|                    | $-\ln L^{sur\ prop}$ | -478.6 | -477.1    | -476.8   |
|                    | $-\ln L^{com\ prop}$ | -495.0 | -494.1    | -494.4   |
|                    | $-\ln L^{prev}$      | -      | -         | -        |
| log priors         | $\ln(k_{ac}^S)$      | -0.54  | -0.65     | -0.65    |
|                    | $move_{y,1}$         | 4.05   | 3.96      | 4.07     |
|                    | $\bar{l}_{1,y}$      | 56.05  | 56.06     | 56.10    |
|                    | $\vartheta_a$        | 0.80   | 0.79      | 0.79     |
|                    | $S_{50}$             | 2.35   | 2.34      | 2.35     |
|                    | $t_{0,c}$            | 0.22   | 0.22      | 0.21     |
|                    | $p_y$                | 1.25   | 1.24      | 1.26     |
|                    | $\varepsilon_y^j$    | -      | 74.44     | 147.54   |
|                    | $k_{ac}^S$           | 0.69   | 0.73      | 0.74     |
|                    | $k_{w,r}$            | 0.30   | 0.32      | 0.32     |
|                    | $k_{s,r}$            | 0.29   | 0.32      | 0.31     |



**Figure 1.** Model  $S_B$  predicted sardine recruitment plotted against spawner biomass for CTS (left) and WTS (right), with the Hockey-Stock (grey dashed) and Beverton Holt (black solid) stock recruitment relationships estimated after conditioning. The three years of peak WTS recruitment are excluded when fitting the stock-recruitment relationship.



**Figure 2.** Model  $S_B$  predicted sardine recruitment plotted against spawner biomass for WTS with the black dots indicating the lowest 6 spawner biomass points, the green dots indicating the highest 4 spawner biomass points, the red dots indicating the 'middle' spawner biomass points and the blue dots indicating the points considered as corresponding to either the 'high' or 'middle' range in the analyses.



**Figure 3.** Model  $S_B$  (top),  $S_{CTS}$  (middle) and  $S_{HS}$  (lower) estimated sardine recruitment plotted against spawner biomass, with the Beverton Holt stock recruitment relationship estimated for CTS during conditioning also shown in the lower two rows and the Hockey Stick stock recruitment relationship estimated for WTS during conditioning shown in the lower row. The colour indicates earlier (lighter) to more recent (darker) years.

**Appendix A: Parameter values and Hessian based SEs**

**Table A1.** Estimated parameter values and hessian based standard errors for all model parameters. Where applicable, upper and lower bounds of parameters are shown. Parameters marked with a \* are likely to be fixed at their joint posterior mode for hybrid MCMC.

| Parameter                               | Year (if applicable) | Min      | Max      | $S_B$    |       | $S_{CTS}$ |       |
|-----------------------------------------|----------------------|----------|----------|----------|-------|-----------|-------|
|                                         |                      |          |          | Estimate | SE    | Estimate  | SE    |
| $N_{W,NL,1983,4,1}^{CTS}$ *             |                      | 0        | 5        | 0.000    | 0.003 | 0.270     | 0.603 |
| $N_{S,NL,1983,4,1}^{WTS}$ *             |                      | 0        | 5        | 0.170    | 0.164 | 0.155     | 0.158 |
| $N_{W,NL,1983,4,1}^{WTS}$ *             |                      | 0        | 5        | 1.790    | 1.848 | 1.387     | 2.077 |
| $R_y^{CTS}/10$ or $\varepsilon_y^{CTS}$ | 1983*                | 0 or -20 | 50 or 20 | 0.000    | 0.000 | -         | -     |
|                                         | 1984                 | 0 or -20 | 50 or 20 | 0.200    | 0.725 | 0.799     | 1.688 |
|                                         | 1985                 | 0 or -20 | 50 or 20 | 1.714    | 0.698 | 1.940     | 1.236 |
|                                         | 1986                 | 0 or -20 | 50 or 20 | 3.160    | 1.270 | 1.323     | 0.550 |
|                                         | 1987                 | 0 or -20 | 50 or 20 | 0.400    | 0.264 | -0.902    | 0.829 |
|                                         | 1988                 | 0 or -20 | 50 or 20 | 0.401    | 0.441 | -0.799    | 1.035 |
|                                         | 1989                 | 0 or -20 | 50 or 20 | 2.098    | 0.976 | 0.745     | 0.680 |
|                                         | 1990                 | 0 or -20 | 50 or 20 | 0.278    | 1.198 | -0.465    | 1.363 |
|                                         | 1991                 | 0 or -20 | 50 or 20 | 2.395    | 1.981 | 0.824     | 1.100 |
|                                         | 1992                 | 0 or -20 | 50 or 20 | 3.320    | 2.459 | 1.193     | 0.936 |
|                                         | 1993                 | 0 or -20 | 50 or 20 | 0.903    | 0.361 | -0.153    | 0.714 |
|                                         | 1994                 | 0 or -20 | 50 or 20 | 4.228    | 1.728 | 1.389     | 0.745 |
|                                         | 1995                 | 0 or -20 | 50 or 20 | 0.860    | 1.221 | -0.112    | 1.239 |
|                                         | 1996                 | 0 or -20 | 50 or 20 | 9.099    | 4.579 | 2.126     | 0.798 |
|                                         | 1997                 | 0 or -20 | 50 or 20 | 1.771    | 2.730 | 0.406     | 1.569 |
|                                         | 1998                 | 0 or -20 | 50 or 20 | 2.215    | 2.159 | 0.494     | 1.345 |
|                                         | 1999                 | 0 or -20 | 50 or 20 | 0.000    | 0.002 | -0.455    | 1.801 |
|                                         | 2000                 | 0 or -20 | 50 or 20 | 0.000    | 0.001 | -0.607    | 1.803 |
|                                         | 2001                 | 0 or -20 | 50 or 20 | 0.000    | 0.000 | -0.974    | 1.621 |
|                                         | 2002                 | 0 or -20 | 50 or 20 | 0.000    | 0.000 | -0.930    | 1.615 |
|                                         | 2003                 | 0 or -20 | 50 or 20 | 0.408    | 0.650 | -0.320    | 1.177 |
|                                         | 2004                 | 0 or -20 | 50 or 20 | 0.000    | 0.002 | -0.516    | 1.548 |
|                                         | 2005                 | 0 or -20 | 50 or 20 | 0.000    | 0.001 | -0.435    | 1.526 |
|                                         | 2006                 | 0 or -20 | 50 or 20 | 0.126    | 0.736 | -0.212    | 1.419 |
|                                         | 2007                 | 0 or -20 | 50 or 20 | 1.348    | 1.961 | 0.667     | 1.858 |
|                                         | 2008                 | 0 or -20 | 50 or 20 | 0.000    | 0.007 | -0.018    | 1.928 |
|                                         | 2009                 | 0 or -20 | 50 or 20 | 2.414    | 0.936 | 1.025     | 0.871 |
|                                         | 2010                 | 0 or -20 | 50 or 20 | 0.639    | 0.545 | -0.273    | 0.901 |
|                                         | 2011                 | 0 or -20 | 50 or 20 | 1.530    | 1.633 | 0.402     | 1.189 |
|                                         | 2012                 | 0 or -20 | 50 or 20 | 0.000    | 0.000 | -1.048    | 1.511 |
|                                         | 2013                 | 0 or -20 | 50 or 20 | 1.336    | 0.453 | 0.314     | 0.627 |
|                                         | 2014                 | 0 or -20 | 50 or 20 | 0.183    | 0.776 | -1.025    | 1.210 |
|                                         | 2015                 | 0 or -20 | 50 or 20 | 0.000    | 0.000 | -1.557    | 1.401 |
|                                         | 2016                 | 0 or -20 | 50 or 20 | 0.000    | 0.000 | -1.984    | 1.277 |
|                                         | 2017                 | 0 or -20 | 50 or 20 | 0.000    | 0.000 | -1.848    | 1.382 |
|                                         | 2018                 | 0 or -20 | 50 or 20 | 0.000    | 0.000 | -1.042    | 1.706 |
|                                         | 2019                 | 0 or -20 | 50 or 20 | 0.000    | 0.000 | -0.062    | 2.405 |
|                                         | 2020                 | 0 or -20 | 50 or 20 | 0.755    | 0.758 | 0.780     | 2.120 |
|                                         | 2021                 | 0 or -20 | 50 or 20 | 2.714    | 1.147 | 1.354     | 0.812 |
|                                         | 2022                 | 0 or -20 | 50 or 20 | 1.331    | 1.675 | 0.251     | 1.299 |
|                                         | 2023                 | 0 or -20 | 50 or 20 | 0.000    | 0.015 | -0.083    | 2.109 |
|                                         | 2024                 | 0 or -20 | 50 or 20 | 1.673    | 4.387 | 0.040     | 2.306 |
| $K^{CTS}/1000$                          |                      | 0        | 10       | -        | -     | 0.231     | 0.146 |
| $h^{CTS}$                               |                      | 0.2      | 1        | -        | -     | 0.548     | 0.380 |
| $R_y^{WTS}/10$                          | 1983*                | 0        | 50       | 0.406    | 0.342 | 0.312     | 0.356 |
|                                         | 1984                 | 0        | 50       | 1.561    | 0.981 | 1.346     | 0.644 |
|                                         | 1985                 | 0        | 50       | 0.052    | 0.141 | 0.055     | 0.138 |
|                                         | 1986                 | 0        | 50       | 0.028    | 0.044 | 0.026     | 0.040 |
|                                         | 1987                 | 0        | 50       | 0.083    | 0.105 | 0.075     | 0.088 |
|                                         | 1988                 | 0        | 50       | 0.738    | 0.414 | 0.677     | 0.378 |

|                    |      |   |    |        |        |        |        |
|--------------------|------|---|----|--------|--------|--------|--------|
|                    | 1989 | 0 | 50 | 0.467  | 0.566  | 0.462  | 0.554  |
|                    | 1990 | 0 | 50 | 1.622  | 1.507  | 1.177  | 1.015  |
|                    | 1991 | 0 | 50 | 1.496  | 2.297  | 1.555  | 2.649  |
|                    | 1992 | 0 | 50 | 5.939  | 4.315  | 5.720  | 4.226  |
|                    | 1993 | 0 | 50 | 0.028  | 0.107  | 0.026  | 0.095  |
|                    | 1994 | 0 | 50 | 7.405  | 4.934  | 6.693  | 4.588  |
|                    | 1995 | 0 | 50 | 3.585  | 2.835  | 3.338  | 2.593  |
|                    | 1996 | 0 | 50 | 6.439  | 6.960  | 6.292  | 6.811  |
|                    | 1997 | 0 | 50 | 7.603  | 4.826  | 7.271  | 4.442  |
|                    | 1998 | 0 | 50 | 6.648  | 3.868  | 6.570  | 3.905  |
|                    | 1999 | 0 | 50 | 16.695 | 6.777  | 14.673 | 6.449  |
|                    | 2000 | 0 | 50 | 39.525 | 11.588 | 36.235 | 10.881 |
|                    | 2001 | 0 | 50 | 39.956 | 11.667 | 36.965 | 10.893 |
|                    | 2002 | 0 | 50 | 35.583 | 9.452  | 32.898 | 8.864  |
|                    | 2003 | 0 | 50 | 3.517  | 1.926  | 3.205  | 1.662  |
|                    | 2004 | 0 | 50 | 2.720  | 1.001  | 2.098  | 1.169  |
|                    | 2005 | 0 | 50 | 6.899  | 2.048  | 5.989  | 2.026  |
|                    | 2006 | 0 | 50 | 2.499  | 1.362  | 2.007  | 1.049  |
|                    | 2007 | 0 | 50 | 2.770  | 2.879  | 3.051  | 2.490  |
|                    | 2008 | 0 | 50 | 5.039  | 1.778  | 3.867  | 2.247  |
|                    | 2009 | 0 | 50 | 10.674 | 3.626  | 10.194 | 3.423  |
|                    | 2010 | 0 | 50 | 2.950  | 1.604  | 2.726  | 1.489  |
|                    | 2011 | 0 | 50 | 3.272  | 2.784  | 3.206  | 2.556  |
|                    | 2012 | 0 | 50 | 6.783  | 2.317  | 5.965  | 2.219  |
|                    | 2013 | 0 | 50 | 0.604  | 0.481  | 0.584  | 0.561  |
|                    | 2014 | 0 | 50 | 2.145  | 1.304  | 1.861  | 0.861  |
|                    | 2015 | 0 | 50 | 1.837  | 0.605  | 1.449  | 0.640  |
|                    | 2016 | 0 | 50 | 3.548  | 1.305  | 3.175  | 1.211  |
|                    | 2017 | 0 | 50 | 1.199  | 0.902  | 0.995  | 0.884  |
|                    | 2018 | 0 | 50 | 2.743  | 1.637  | 2.409  | 1.540  |
|                    | 2019 | 0 | 50 | 8.346  | 3.414  | 7.446  | 3.195  |
|                    | 2020 | 0 | 50 | 1.284  | 1.317  | 1.127  | 1.690  |
|                    | 2021 | 0 | 50 | 2.668  | 1.292  | 2.494  | 1.202  |
|                    | 2022 | 0 | 50 | 3.128  | 2.514  | 3.014  | 2.355  |
|                    | 2023 | 0 | 50 | 5.926  | 2.024  | 4.400  | 3.065  |
|                    | 2024 | 0 | 50 | 0.497  | 3.563  | 1.124  | 4.520  |
| $move_{y,0}^{WTS}$ | 1984 | 0 | 1  | 0.015  | 0.039  | 0.019  | 0.053  |
|                    | 1985 | 0 | 1  | 0.793  | 0.493  | 0.911  | 0.306  |
|                    | 1986 | 0 | 1  | 0.479  | 1.754  | 0.568  | 1.665  |
|                    | 1987 | 0 | 1  | 0.780  | 0.874  | 0.790  | 0.835  |
|                    | 1988 | 0 | 1  | 0.759  | 0.867  | 0.785  | 0.788  |
|                    | 1989 | 0 | 1  | 0.944  | 0.242  | 0.953  | 0.203  |
|                    | 1990 | 0 | 1  | 0.340  | 1.728  | 0.344  | 1.821  |
|                    | 1991 | 0 | 1  | 0.361  | 0.471  | 0.480  | 0.588  |
|                    | 1992 | 0 | 1  | 0.615  | 0.983  | 0.574  | 1.084  |
|                    | 1993 | 0 | 1  | 0.157  | 0.174  | 0.154  | 0.171  |
|                    | 1994 | 0 | 1  | 0.702  | 0.711  | 0.666  | 0.722  |
|                    | 1995 | 0 | 1  | 0.962  | 0.164  | 0.957  | 0.185  |
|                    | 1996 | 0 | 1  | 0.869  | 0.560  | 0.869  | 0.565  |
|                    | 1997 | 0 | 1  | 0.290  | 0.508  | 0.275  | 0.481  |
|                    | 1998 | 0 | 1  | 0.707  | 0.472  | 0.686  | 0.465  |
|                    | 1999 | 0 | 1  | 0.673  | 1.027  | 0.569  | 1.008  |
|                    | 2000 | 0 | 1  | 0.435  | 0.396  | 0.492  | 0.428  |
|                    | 2001 | 0 | 1  | 0.666  | 0.129  | 0.690  | 0.136  |
|                    | 2002 | 0 | 1  | 0.520  | 0.152  | 0.538  | 0.157  |
|                    | 2003 | 0 | 1  | 0.340  | 0.194  | 0.353  | 0.199  |
|                    | 2004 | 0 | 1  | 0.609  | 0.187  | 0.645  | 0.190  |
|                    | 2005 | 0 | 1  | 0.516  | 0.322  | 0.710  | 0.512  |
|                    | 2006 | 0 | 1  | 0.265  | 0.284  | 0.358  | 0.335  |
|                    | 2007 | 0 | 1  | 0.619  | 0.310  | 0.791  | 0.416  |
|                    | 2008 | 0 | 1  | 0.193  | 0.585  | 0.157  | 0.447  |
|                    | 2009 | 0 | 1  | 0.065  | 0.221  | 0.093  | 0.310  |

|                  |       |   |   |       |       |       |       |
|------------------|-------|---|---|-------|-------|-------|-------|
|                  | 2010  | 0 | 1 | 0.349 | 0.212 | 0.363 | 0.222 |
|                  | 2011  | 0 | 1 | 0.982 | 0.080 | 0.982 | 0.078 |
|                  | 2012  | 0 | 1 | 0.544 | 0.610 | 0.524 | 0.567 |
|                  | 2013  | 0 | 1 | 0.235 | 0.184 | 0.252 | 0.199 |
|                  | 2014  | 0 | 1 | 0.542 | 0.433 | 0.610 | 0.489 |
|                  | 2015  | 0 | 1 | 0.342 | 1.428 | 0.488 | 1.540 |
|                  | 2016  | 0 | 1 | 0.010 | 0.044 | 0.013 | 0.058 |
|                  | 2017  | 0 | 1 | 0.507 | 0.208 | 0.577 | 0.222 |
|                  | 2018  | 0 | 1 | 0.681 | 0.202 | 0.781 | 0.218 |
|                  | 2019  | 0 | 1 | 0.743 | 0.223 | 0.780 | 0.251 |
|                  | 2020  | 0 | 1 | 0.834 | 0.144 | 0.863 | 0.236 |
|                  | 2021  | 0 | 1 | 0.979 | 0.103 | 0.817 | 1.219 |
|                  | 2022  | 0 | 1 | 0.488 | 1.592 | 0.606 | 1.709 |
|                  | 2023  | 0 | 1 | 0.802 | 0.753 | 0.784 | 0.780 |
|                  | 2024  | 0 | 1 | 0.060 | 0.244 | 0.097 | 0.423 |
| $p_y$            | 1983* | 0 | 1 | 1.000 | 0.000 | 1.000 | 0.000 |
|                  | 1984  | 0 | 1 | 0.995 | 0.009 | 0.994 | 0.009 |
|                  | 1985  | 0 | 1 | 0.130 | 1.348 | 0.139 | 1.385 |
|                  | 1986  | 0 | 1 | 0.652 | 1.117 | 0.671 | 1.071 |
|                  | 1987  | 0 | 1 | 0.929 | 0.236 | 0.930 | 0.231 |
|                  | 1988  | 0 | 1 | 0.999 | 0.014 | 0.999 | 0.014 |
|                  | 1989  | 0 | 1 | 0.075 | 0.860 | 0.077 | 0.916 |
|                  | 1990  | 0 | 1 | 0.968 | 0.115 | 0.955 | 0.161 |
|                  | 1991  | 0 | 1 | 0.997 | 0.029 | 0.997 | 0.029 |
|                  | 1992  | 0 | 1 | 0.999 | 0.008 | 0.999 | 0.007 |
|                  | 1993  | 0 | 1 | 0.337 | 2.475 | 0.326 | 2.427 |
|                  | 1994  | 0 | 1 | 0.980 | 0.016 | 0.980 | 0.016 |
|                  | 1995  | 0 | 1 | 0.240 | 0.425 | 0.224 | 0.397 |
|                  | 1996  | 0 | 1 | 0.998 | 0.019 | 0.998 | 0.018 |
|                  | 1997  | 0 | 1 | 0.982 | 0.015 | 0.982 | 0.015 |
|                  | 1998  | 0 | 1 | 0.257 | 0.379 | 0.299 | 0.356 |
|                  | 1999  | 0 | 1 | 0.847 | 0.087 | 0.838 | 0.096 |
|                  | 2000  | 0 | 1 | 0.999 | 0.001 | 0.999 | 0.001 |
|                  | 2001  | 0 | 1 | 0.926 | 0.039 | 0.926 | 0.039 |
|                  | 2002  | 0 | 1 | 0.944 | 0.027 | 0.944 | 0.027 |
|                  | 2003  | 0 | 1 | 0.871 | 0.105 | 0.868 | 0.105 |
|                  | 2004  | 0 | 1 | 0.781 | 0.103 | 0.735 | 0.160 |
|                  | 2005  | 0 | 1 | 0.668 | 0.132 | 0.647 | 0.146 |
|                  | 2006  | 0 | 1 | 0.722 | 0.200 | 0.681 | 0.212 |
|                  | 2007  | 0 | 1 | 0.571 | 0.490 | 0.641 | 0.343 |
|                  | 2008  | 0 | 1 | 0.643 | 0.157 | 0.566 | 0.265 |
|                  | 2009  | 0 | 1 | 0.957 | 0.029 | 0.959 | 0.028 |
|                  | 2010  | 0 | 1 | 0.953 | 0.192 | 0.951 | 0.197 |
|                  | 2011  | 0 | 1 | 0.934 | 0.215 | 0.934 | 0.208 |
|                  | 2012  | 0 | 1 | 1.000 | 0.000 | 1.000 | 0.000 |
|                  | 2013  | 0 | 1 | 0.057 | 0.522 | 0.093 | 0.746 |
|                  | 2014  | 0 | 1 | 0.134 | 0.388 | 0.060 | 0.225 |
|                  | 2015  | 0 | 1 | 0.744 | 0.129 | 0.697 | 0.167 |
|                  | 2016  | 0 | 1 | 0.791 | 0.100 | 0.785 | 0.104 |
|                  | 2017  | 0 | 1 | 0.934 | 0.228 | 0.936 | 0.241 |
|                  | 2018  | 0 | 1 | 0.857 | 0.240 | 0.852 | 0.256 |
|                  | 2019  | 0 | 1 | 0.709 | 0.187 | 0.695 | 0.196 |
|                  | 2020  | 0 | 1 | 0.955 | 0.050 | 0.952 | 0.075 |
|                  | 2021  | 0 | 1 | 0.018 | 0.177 | 0.018 | 0.179 |
|                  | 2022  | 0 | 1 | 0.672 | 0.377 | 0.675 | 0.363 |
|                  | 2023  | 0 | 1 | 0.530 | 0.147 | 0.409 | 0.383 |
|                  | 2024  | 0 | 1 | 0.494 | 3.490 | 0.781 | 0.911 |
| $\chi_{w,y,2}^*$ | 1984  | 0 | 1 | 0.393 | 0.773 | 0.471 | 0.835 |
|                  | 1985  | 0 | 1 | 0.105 | 0.309 | 0.106 | 0.318 |
|                  | 1986  | 0 | 1 | 0.210 | 0.215 | 0.227 | 0.234 |
|                  | 1987  | 0 | 1 | 0.012 | 0.048 | 0.013 | 0.050 |
|                  | 1988  | 0 | 1 | 0.070 | 0.308 | 0.067 | 0.299 |



|                  |      |   |   |       |       |       |       |
|------------------|------|---|---|-------|-------|-------|-------|
|                  | 1989 | 0 | 1 | 0.856 | 0.576 | 0.841 | 0.624 |
|                  | 1990 | 0 | 1 | 0.326 | 0.749 | 0.324 | 0.737 |
|                  | 1991 | 0 | 1 | 0.067 | 0.188 | 0.069 | 0.194 |
|                  | 1992 | 0 | 1 | 0.022 | 0.090 | 0.021 | 0.088 |
|                  | 1993 | 0 | 1 | 0.005 | 0.023 | 0.006 | 0.026 |
|                  | 1994 | 0 | 1 | 0.090 | 0.276 | 0.092 | 0.286 |
|                  | 1995 | 0 | 1 | 0.028 | 0.115 | 0.030 | 0.123 |
|                  | 1996 | 0 | 1 | 0.660 | 0.872 | 0.669 | 0.880 |
|                  | 1997 | 0 | 1 | 0.041 | 0.132 | 0.044 | 0.140 |
|                  | 1998 | 0 | 1 | 0.041 | 0.141 | 0.045 | 0.156 |
|                  | 1999 | 0 | 1 | 0.302 | 0.823 | 0.310 | 0.852 |
|                  | 2000 | 0 | 1 | 0.022 | 0.098 | 0.023 | 0.103 |
|                  | 2001 | 0 | 1 | 0.009 | 0.038 | 0.009 | 0.039 |
|                  | 2002 | 0 | 1 | 0.038 | 0.102 | 0.041 | 0.109 |
|                  | 2003 | 0 | 1 | 0.057 | 0.146 | 0.061 | 0.157 |
|                  | 2004 | 0 | 1 | 0.058 | 0.208 | 0.059 | 0.212 |
|                  | 2005 | 0 | 1 | 0.092 | 0.348 | 0.090 | 0.343 |
|                  | 2006 | 0 | 1 | 0.025 | 0.080 | 0.027 | 0.088 |
|                  | 2007 | 0 | 1 | 0.015 | 0.062 | 0.016 | 0.068 |
|                  | 2008 | 0 | 1 | 0.008 | 0.035 | 0.009 | 0.041 |
|                  | 2009 | 0 | 1 | 0.018 | 0.068 | 0.023 | 0.088 |
|                  | 2010 | 0 | 1 | 0.042 | 0.117 | 0.042 | 0.120 |
|                  | 2011 | 0 | 1 | 0.052 | 0.188 | 0.059 | 0.215 |
|                  | 2012 | 0 | 1 | 0.026 | 0.116 | 0.028 | 0.126 |
|                  | 2013 | 0 | 1 | 0.014 | 0.062 | 0.016 | 0.068 |
|                  | 2014 | 0 | 1 | 0.017 | 0.074 | 0.017 | 0.075 |
|                  | 2015 | 0 | 1 | 0.056 | 0.233 | 0.062 | 0.259 |
|                  | 2016 | 0 | 1 | 0.051 | 0.220 | 0.055 | 0.237 |
|                  | 2017 | 0 | 1 | 0.011 | 0.051 | 0.012 | 0.055 |
|                  | 2018 | 0 | 1 | 0.029 | 0.116 | 0.031 | 0.124 |
|                  | 2019 | 0 | 1 | 0.006 | 0.029 | 0.007 | 0.033 |
|                  | 2020 | 0 | 1 | 0.003 | 0.014 | 0.004 | 0.018 |
|                  | 2021 | 0 | 1 | 0.010 | 0.046 | 0.015 | 0.067 |
|                  | 2022 | 0 | 1 | 0.005 | 0.025 | 0.013 | 0.061 |
|                  | 2023 | 0 | 1 | 0.025 | 0.107 | 0.028 | 0.122 |
|                  | 2024 | 0 | 1 | 0.060 | 0.212 | 0.062 | 0.223 |
| $\chi_{w,y,3}^*$ | 1984 | 0 | 1 | 0.246 | 0.894 | 0.253 | 0.937 |
|                  | 1985 | 0 | 1 | 0.387 | 1.315 | 0.308 | 0.160 |
|                  | 1986 | 0 | 1 | 0.334 | 1.304 | 0.359 | 1.332 |
|                  | 1987 | 0 | 1 | 0.001 | 0.004 | 0.001 | 0.004 |
|                  | 1988 | 0 | 1 | 0.094 | 0.256 | 0.083 | 0.228 |
|                  | 1989 | 0 | 1 | 0.069 | 0.171 | 0.057 | 0.144 |
|                  | 1990 | 0 | 1 | 0.006 | 0.024 | 0.006 | 0.024 |
|                  | 1991 | 0 | 1 | 0.012 | 0.039 | 0.011 | 0.038 |
|                  | 1992 | 0 | 1 | 0.194 | 0.339 | 0.188 | 0.337 |
|                  | 1993 | 0 | 1 | 0.055 | 0.095 | 0.057 | 0.098 |
|                  | 1994 | 0 | 1 | 0.752 | 0.860 | 0.754 | 0.861 |
|                  | 1995 | 0 | 1 | 0.542 | 0.807 | 0.561 | 0.827 |
|                  | 1996 | 0 | 1 | 0.349 | 0.764 | 0.354 | 0.779 |
|                  | 1997 | 0 | 1 | 0.032 | 0.072 | 0.034 | 0.078 |
|                  | 1998 | 0 | 1 | 0.266 | 0.521 | 0.286 | 0.565 |
|                  | 1999 | 0 | 1 | 0.095 | 0.277 | 0.099 | 0.291 |
|                  | 2000 | 0 | 1 | 0.019 | 0.061 | 0.019 | 0.062 |
|                  | 2001 | 0 | 1 | 0.018 | 0.044 | 0.017 | 0.045 |
|                  | 2002 | 0 | 1 | 0.051 | 0.095 | 0.053 | 0.101 |
|                  | 2003 | 0 | 1 | 0.034 | 0.081 | 0.035 | 0.084 |
|                  | 2004 | 0 | 1 | 0.136 | 0.422 | 0.138 | 0.431 |
|                  | 2005 | 0 | 1 | 0.015 | 0.059 | 0.016 | 0.066 |
|                  | 2006 | 0 | 1 | 0.011 | 0.030 | 0.014 | 0.036 |
|                  | 2007 | 0 | 1 | 0.010 | 0.043 | 0.011 | 0.048 |
|                  | 2008 | 0 | 1 | 0.004 | 0.015 | 0.005 | 0.019 |
|                  | 2009 | 0 | 1 | 0.008 | 0.035 | 0.009 | 0.038 |

|                                     |           |   |    |        |       |        |       |
|-------------------------------------|-----------|---|----|--------|-------|--------|-------|
|                                     | 2010      | 0 | 1  | 0.048  | 0.086 | 0.052  | 0.096 |
|                                     | 2011      | 0 | 1  | 0.850  | 0.581 | 0.852  | 0.576 |
|                                     | 2012      | 0 | 1  | 0.016  | 0.051 | 0.017  | 0.054 |
|                                     | 2013      | 0 | 1  | 0.005  | 0.024 | 0.006  | 0.025 |
|                                     | 2014      | 0 | 1  | 0.016  | 0.072 | 0.018  | 0.081 |
|                                     | 2015      | 0 | 1  | 0.048  | 0.215 | 0.049  | 0.223 |
|                                     | 2016      | 0 | 1  | 0.036  | 0.101 | 0.040  | 0.112 |
|                                     | 2017      | 0 | 1  | 0.003  | 0.015 | 0.004  | 0.018 |
|                                     | 2018      | 0 | 1  | 0.009  | 0.041 | 0.010  | 0.047 |
|                                     | 2019      | 0 | 1  | 0.033  | 0.070 | 0.044  | 0.097 |
|                                     | 2020      | 0 | 1  | 0.002  | 0.008 | 0.002  | 0.011 |
|                                     | 2021      | 0 | 1  | 0.249  | 0.452 | 0.579  | 1.019 |
|                                     | 2022      | 0 | 1  | 0.017  | 0.053 | 0.025  | 0.077 |
|                                     | 2023      | 0 | 1  | 0.327  | 0.580 | 0.333  | 0.600 |
|                                     | 2024      | 0 | 1  | 0.052  | 0.151 | 0.052  | 0.155 |
| $\bar{l}_{1,y}$                     | 1984      |   |    | 9.683  | 1.012 | 9.704  | 1.019 |
|                                     | 1985      |   |    | 9.389  | 0.943 | 9.408  | 0.933 |
|                                     | 1986      |   |    | 9.795  | 0.880 | 9.798  | 0.885 |
|                                     | 1987      |   |    | 10.207 | 0.896 | 10.159 | 0.888 |
|                                     | 1988      |   |    | 10.165 | 0.877 | 10.163 | 0.879 |
|                                     | 1989      |   |    | 11.219 | 0.775 | 11.211 | 0.763 |
|                                     | 1990      |   |    | 10.344 | 0.865 | 10.355 | 0.855 |
|                                     | 1991      |   |    | 9.947  | 0.764 | 9.945  | 0.765 |
|                                     | 1992      |   |    | 10.263 | 0.695 | 10.263 | 0.694 |
|                                     | 1993      |   |    | 10.043 | 0.729 | 10.061 | 0.731 |
|                                     | 1994      |   |    | 10.394 | 0.778 | 10.434 | 0.776 |
|                                     | 1995      |   |    | 9.842  | 0.750 | 9.851  | 0.753 |
|                                     | 1996      |   |    | 10.164 | 0.797 | 10.142 | 0.799 |
|                                     | 1997      |   |    | 9.773  | 0.753 | 9.788  | 0.756 |
|                                     | 1998      |   |    | 10.184 | 0.767 | 10.160 | 0.774 |
|                                     | 1999      |   |    | 10.135 | 0.707 | 10.136 | 0.705 |
|                                     | 2000      |   |    | 10.089 | 0.721 | 10.104 | 0.723 |
|                                     | 2001      |   |    | 10.392 | 0.687 | 10.433 | 0.690 |
|                                     | 2002      |   |    | 10.554 | 0.720 | 10.585 | 0.720 |
|                                     | 2003      |   |    | 10.038 | 0.810 | 10.072 | 0.800 |
|                                     | 2004      |   |    | 9.982  | 0.807 | 10.053 | 0.802 |
|                                     | 2005      |   |    | 9.905  | 0.802 | 9.942  | 0.806 |
|                                     | 2006      |   |    | 10.475 | 0.795 | 10.478 | 0.794 |
|                                     | 2007      |   |    | 10.090 | 0.758 | 10.161 | 0.756 |
|                                     | 2008      |   |    | 9.823  | 0.772 | 9.794  | 0.779 |
|                                     | 2009      |   |    | 10.135 | 0.718 | 10.106 | 0.714 |
|                                     | 2010      |   |    | 10.321 | 0.728 | 10.347 | 0.727 |
|                                     | 2011      |   |    | 9.579  | 0.683 | 9.576  | 0.682 |
|                                     | 2012      |   |    | 10.169 | 0.730 | 10.188 | 0.727 |
|                                     | 2013      |   |    | 10.279 | 0.773 | 10.276 | 0.774 |
|                                     | 2014      |   |    | 10.901 | 0.762 | 10.943 | 0.766 |
|                                     | 2015      |   |    | 10.533 | 0.846 | 10.545 | 0.851 |
|                                     | 2016      |   |    | 10.648 | 0.817 | 10.686 | 0.816 |
|                                     | 2017      |   |    | 10.249 | 0.746 | 10.263 | 0.750 |
|                                     | 2018      |   |    | 9.559  | 0.700 | 9.579  | 0.702 |
|                                     | 2019      |   |    | 9.588  | 0.950 | 9.528  | 0.954 |
|                                     | 2020      |   |    | 8.554  | 0.663 | 8.488  | 0.661 |
|                                     | 2021      |   |    | 9.517  | 0.724 | 9.435  | 0.718 |
|                                     | 2022      |   |    | 9.835  | 0.692 | 9.619  | 0.703 |
|                                     | 2023      |   |    | 9.164  | 0.798 | 9.157  | 0.795 |
|                                     | 2024      |   |    | 9.815  | 0.744 | 9.848  | 0.738 |
| $(\sigma_1^{sel})^2$                |           | 2 | 15 | 3.782  | 2.045 | 3.730  | 2.030 |
| $(\sigma_2^{sel})^2$                |           | 0 | 10 | 0.724  | 0.088 | 0.717  | 0.089 |
| $\bar{l}_{2,s,y,1} - \bar{l}_{1,y}$ | 1984-1997 | 0 | 15 | 9.992  | 0.987 | 9.955  | 0.974 |
| $\bar{l}_{2,s,y,3} - \bar{l}_{1,y}$ | 1984-1997 | 0 | 15 | 10.145 | 0.932 | 10.107 | 0.919 |
| $\bar{l}_{2,w,y,1} - \bar{l}_{1,y}$ | 1984-1993 | 0 | 15 | 8.218  | 1.345 | 8.259  | 1.401 |

|                                     |           |       |          |        |       |        |       |
|-------------------------------------|-----------|-------|----------|--------|-------|--------|-------|
| $\bar{l}_{2,w,y,3} - \bar{l}_{1,y}$ | 1984-1993 | 0     | 15       | 9.356  | 0.943 | 9.382  | 0.949 |
| $\bar{l}_{2,s,y,1} - \bar{l}_{1,y}$ | 1998-2024 | 0     | 15       | 6.067  | 0.451 | 6.081  | 0.445 |
| $\bar{l}_{2,s,y,3} - \bar{l}_{1,y}$ | 1998-2024 | 0     | 15       | 6.917  | 0.532 | 6.926  | 0.523 |
| $\bar{l}_{2,w,y,1} - \bar{l}_{1,y}$ | 1994-2024 | 0     | 15       | 6.529  | 0.498 | 6.408  | 0.495 |
| $\bar{l}_{2,w,y,3} - \bar{l}_{1,y}$ | 1994-2024 | 0     | 15       | 6.088  | 0.577 | 5.927  | 0.596 |
| $L_{w,1}$                           |           | 5     | 18       | 12.487 | 0.230 | 12.472 | 0.235 |
| $L_{s,1} - L_{w,1}$                 |           | 0     | 13       | 2.004  | 0.574 | 2.040  | 0.580 |
| $\kappa_w$                          |           | 0     | 2        | 1.140  | 0.075 | 1.139  | 0.077 |
| $\kappa_s - \kappa_w$               |           | 0     | 2        | 0.000  | 0.000 | 0.000  | 0.000 |
| $t_{0,s}$                           |           | -0.25 | 0.25     | -0.210 | 0.086 | -0.209 | 0.086 |
| $t_{0,w}$                           |           | -0.25 | 0.25     | 0.144  | 0.042 | 0.142  | 0.044 |
| $\vartheta_0$                       |           | 0     | 3        | 2.954  | 0.198 | 2.950  | 0.217 |
| $\vartheta_1$                       |           | 0     | 3        | 2.187  | 0.206 | 2.158  | 0.208 |
| $\vartheta_{2^+}$                   |           | 0     | 3        | 1.137  | 0.128 | 1.128  | 0.128 |
| $\vartheta_{bf}$                    |           | 0     | 1        | 0.009  | 0.040 | 0.009  | 0.041 |
| $\vartheta_{af}$                    |           | 0     | 1        | 0.017  | 0.074 | 0.018  | 0.076 |
| $k_{ac}$                            |           | 0     | $\infty$ | 0.688  | 0.133 | 0.734  | 0.141 |
| $k_{cov}$                           |           | 0     | 1        | 0.432  | 0.046 | 0.434  | 0.046 |
| $k_{covS}$                          |           | 0     | 1        | 0.987  | 0.059 | 0.987  | 0.059 |
| $S_{50}$                            |           |       |          | 8.124  | 0.585 | 8.120  | 0.582 |
| $\delta$                            |           |       |          | 0.737  | 0.205 | 0.731  | 0.206 |